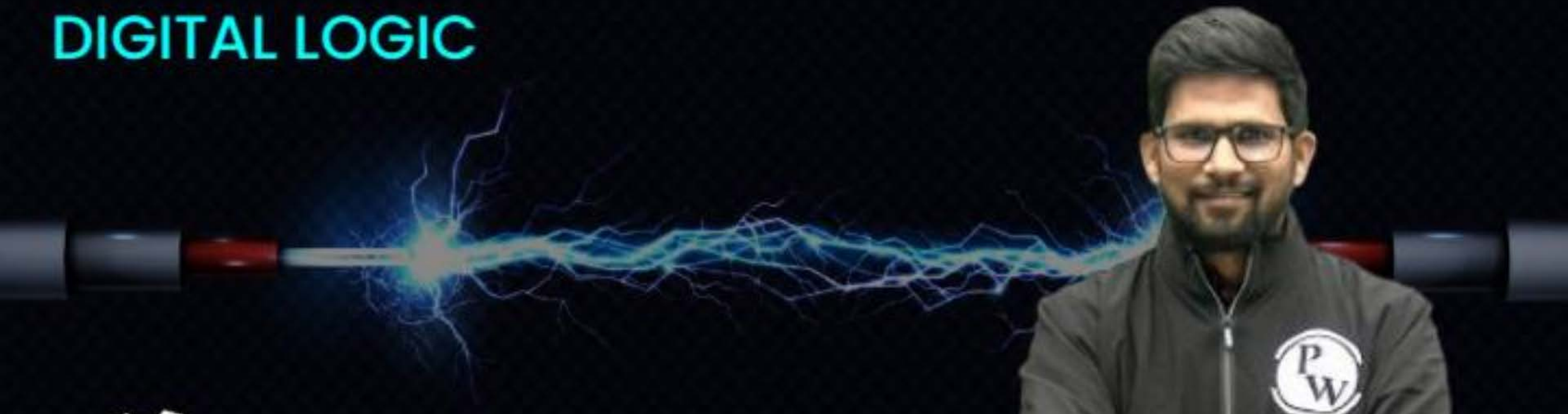
COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No: 01

Sequential Circuit



By- Chandan Gupta Sir

Recap of Previous Lecture



- Question Practice
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- S-R latch
- S-R ff

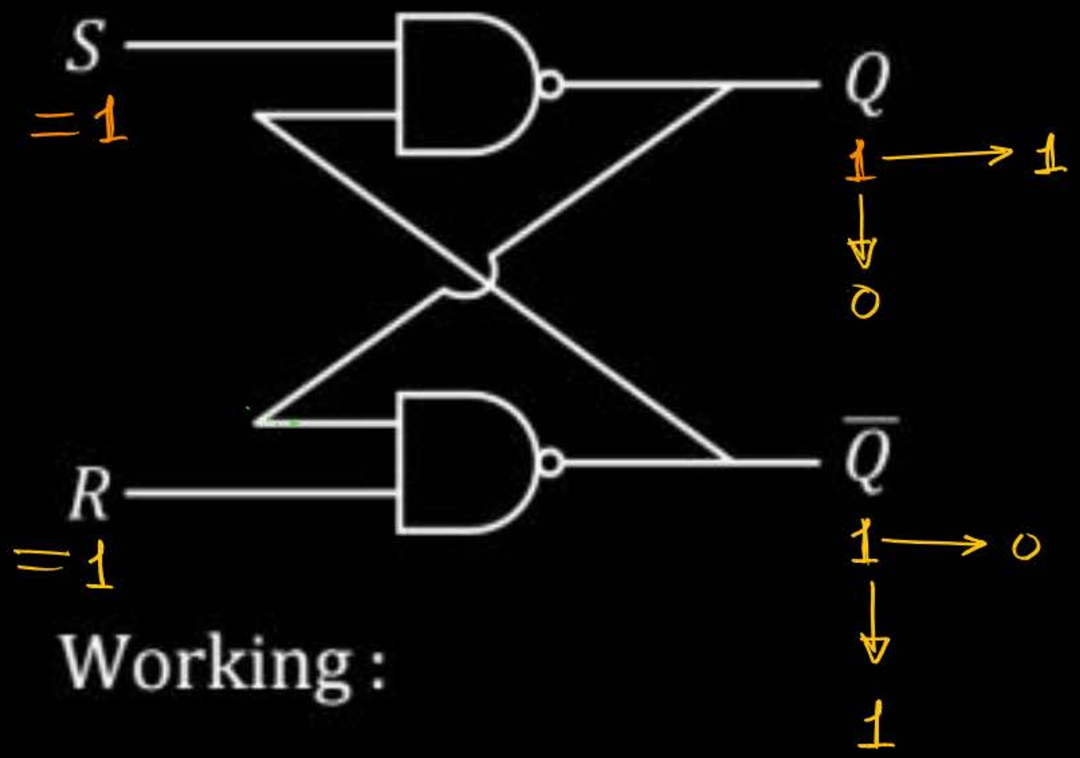
Sequential Circuit

- Basic element of sequential circuit ?

  ↳ S-R latch

S-R Latch :

SR latch using NAND gate :



→ Working :

S	R	Q(n+1)
0	0	invalid
0	1	1
1	0	0
1	1	Q(n) hold state.

S=0 & R=0 is invalid i/p condition as it generates invalid o/p states [Q & $\overline{Q}$ coming out to be same.].

$S=0, R=1$ $Q(n)=?$ $Q(n+1)=1$

$S=1, R=1$ $Q(n)=1$ $Q(n+1)=1$

$S=1, R=0$ $Q(n)=1$ $Q(n+1)=0$

$S=1, R=1$ $Q(n)=0$ $Q(n+1)=0$

$S=0, R=0$ $Q(n)=0$ $Q(n+1)=1$, $\overline{Q}(n+1)=1$

$S=0, R=1$ $Q(n)=1$, $\overline{Q}(n)=1$ $Q(n+1)=1$

$S=0, R=0$ $Q(n)=1$ $Q(n+1)=1$, $\overline{Q}(n+1)=1$

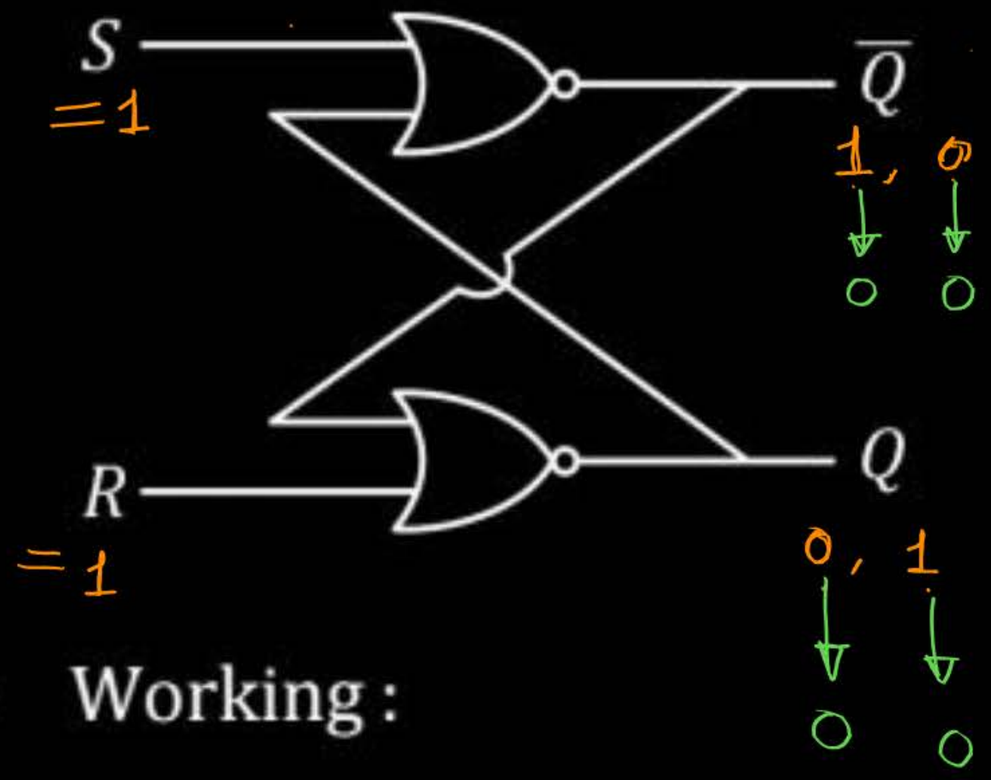
$S=1, R=0$ $Q(n)=1$, $\overline{Q}(n)=1$ $Q(n+1)=0$

$S=0, R=0$ $Q(n)=0$ $Q(n+1)=1$, $\overline{Q}(n+1)=1$

$S=1, R=1$, $Q(n)=1$, $\overline{Q}(n)=1$ will not hold the invalid o/p state

and will go to any of the valid o/p state $(0,1)$ or $(1,0)$.

SR latch using NOR gate :



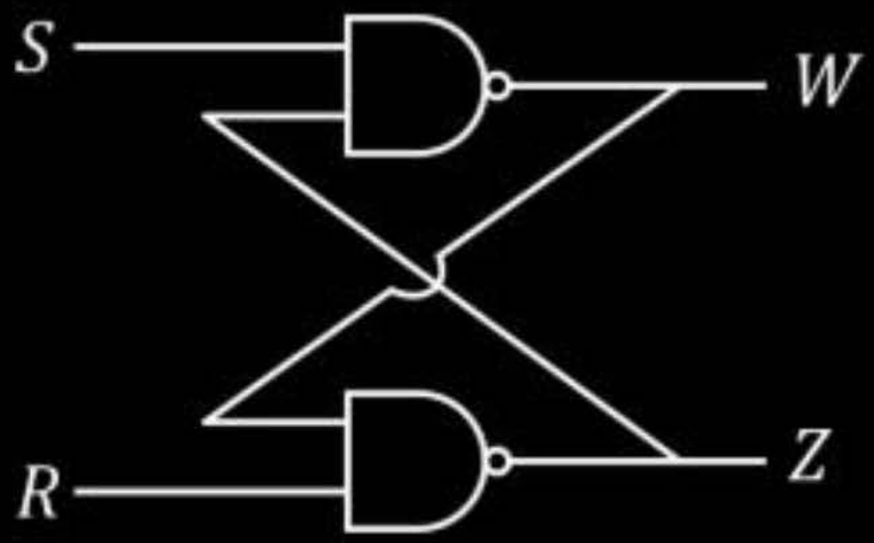
S	R	Q(n+1)
0	0	Q(n)
0	1	0
1	0	1
1	1	invalid O/P

→ Working :

S=1, R=1 is invalid i/p condition as it generates invalid o/ps (Q & $\overline{Q}$ coming out to be same)

Question

A NAND latch circuit is given below :



$S=0, R=0$ $W=?$ $Z=?$ $W^+=1$ $Z^+=1$

$S=1, R=1$ $W=1$ $Z=1$ $W^+=1, Z^+=0$ or $W^+=0, Z^+=1$

Initially input $S = R = 0$ and after sometime input changes from $S = R = 0$ to $S = R = 1$ then

(a) Output changes from $W = Z = 1$ to $W = Z = 1$.

(b) Output changes from $W = Z = 1$ to either W = 1 and Z = 0 or W = 0 and Z = 1

(c) Output changes from $W = Z = 1$ to $W = Z = 0$

(d) None of these

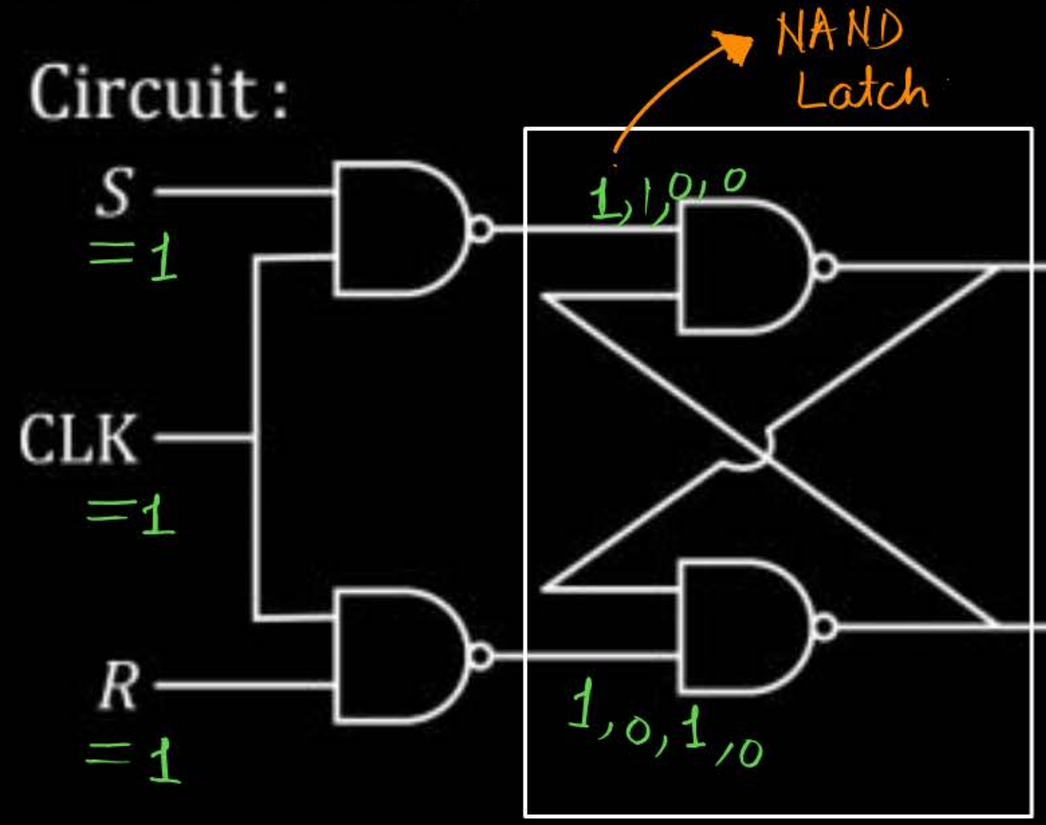
- IMP Points :

- Corresponding to hold state – i/p , Latch holds the present O/P state in next O/P state only if the present O/P state is valid O/P state.

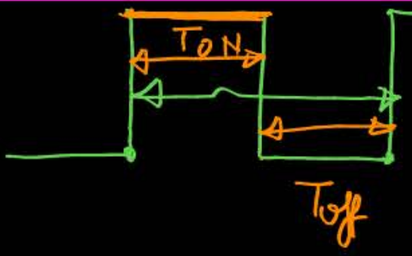
  → if present O/P state is invalid O/P state then it will not hold the invalid O/P state in next O/P state rather it will go any of the valid O/P state (1, 0) or (0, 1).

S-R Flip-Flop

- Circuit :



Positive level triggered ff



$T_{CLK} = T_{ON} + T_{off}$

% duty cycle $= \frac{T_{ON}}{T_{ON}+T_{off}} \times 100\%$

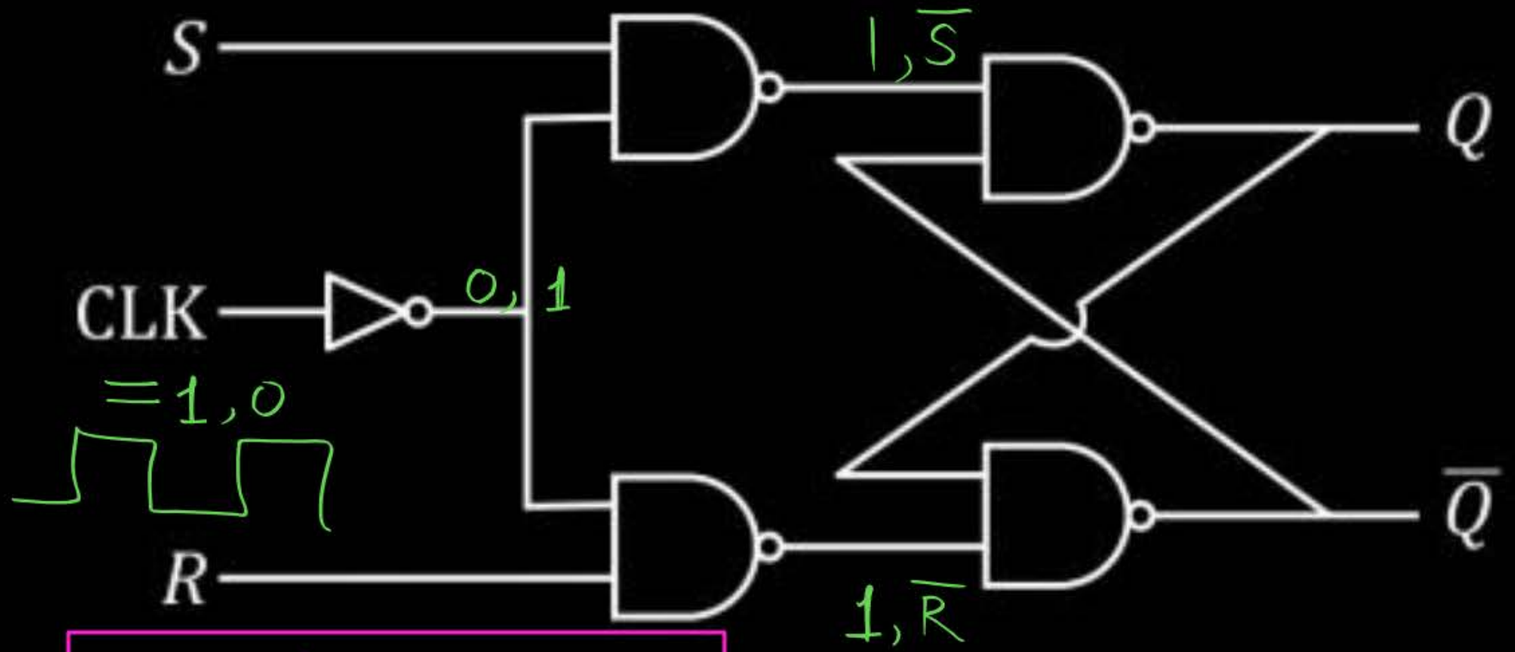
if $T_{ON} = T_{off}$ % duty cycle = 50%

if $T_{ON} > T_{off}$ ⟶ % duty cycle > 50%

$= \frac{1}{1+\frac{T_{off}}{T_{ON}}} \times 100\%$

if $T_{ON} < T_{off}$ ⟶ % duty cycle < 50%

If circuit is changed to :



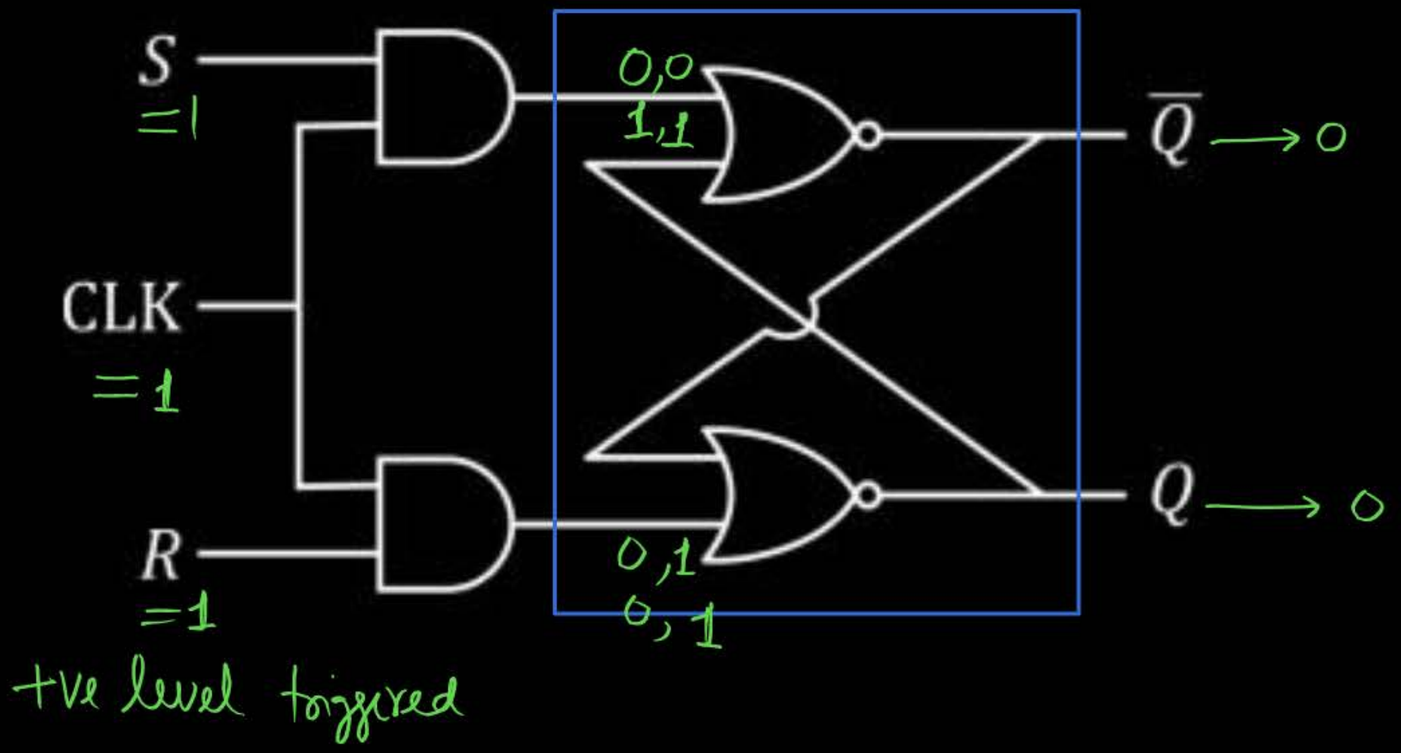
$= 1, 0$

Negative level triggered ff.

- Truth Table :

CLK	S	R	$Q(n+1)$
Non-Active	X	X	Q(n)
Active	0	0	Q(n)
Active	0	1	0
Active	1	0	1
Active	1	1	invalid o/p state.

S-R FF – using NOR latch + Clock



Then truth table will be :

CLK	S	R	$Q(n+1)$
Non Active	X	X	$Q(n)$
Active	0	0	$Q(n)$
Active	0	1	0
Active	1	0	1
Active	1	1	invalid 0/? state



Topic : 2 Min Summary

S-R latch
S-R ff

Thank you

GW

Soldiers !

